10.5 Angle Relationships in Circles

Name _____ Period ____ Date _____

Geometry 10.5 · Student sheet

I can…

- find the angle formed by a tangent and a chord.
- find angles formed by lines that meet inside or outside a circle.
- find the measure of a circumscribed angle.

Vocabulary · fill in the missing word

Circumscribed angle — an angle whose sides are _____ to a circle

Secant — a line that _____ a circle in two points

Tangent — a line in the plane of a circle that _____ the circle in exactly one point

Part 1 · Discovery: vertex inside the circle

Draw a circle with center O and two chords that cross inside it. Measure one angle where they cross, and the two arcs that angle and its vertical angle open onto.

Angle	Arc 1	Arc 2	Arc 1 + Arc 2	½ (Arc 1 + Arc 2)

What do you notice?

Part 2 · Notes and examples (fill in each note, then work the example)

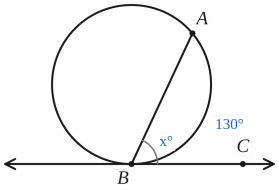
Tangent and Intersected Chord Theorem: the angle formed by a tangent and a chord at the point of tangency is _____ its intercepted arc.

Example 1

Line BC is tangent to the circle at B.

a. Find m∠ABC.

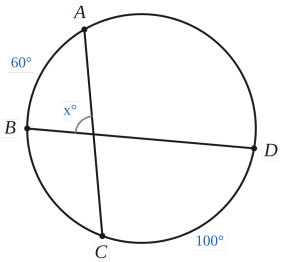
b. If the angle formed by a tangent and a chord is 104°, what is its intercepted arc?



Angles Inside the Circle Theorem: when two chords cross inside a circle, each angle is one-half the _____ of the arcs intercepted by the angle and its vertical angle.

Example 2

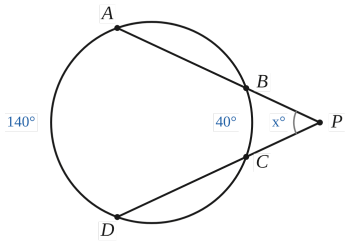
Two chords cross inside the circle. Find the value of x.



Angles Outside the Circle Theorem: when tangents or secants meet outside a circle, the angle is one-half the _____ of the intercepted arcs (far arc minus near arc).

Example 3

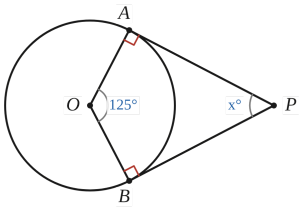
Two secants meet at P outside the circle. Find the value of x.



Circumscribed Angle Theorem: a circumscribed angle measures 180° minus the _____ angle that intercepts the same arc.

Example 4

$\overline{PA}$ and $\overline{PB}$ are tangent to circle O. Find m∠APB.



Example 5

From a satellite, the two lines of sight to the horizon are tangent to Earth. The arc of Earth you can see measures 166°. What is the angle between the lines of sight?

Part 3 · Partner practice

Partner 1

Two chords cross inside a circle. The arcs intercepted by an angle and its vertical angle measure 84° and 116°. Find the angle.

Partner 2

A tangent and a secant meet outside a circle. The far arc measures 160° and the near arc measures 50°. Find the angle.